

NIIS HACKATHON 2026 — OFFICIAL PROBLEM BRIEF

Track: Campus Nexus

Challenge ID: PS-03

Title: Sustainable Campus & Facility Intelligence Dashboard

1. Executive Summary & Problem Overview

Higher educational institutions consume massive volumes of electrical power, municipal water, and daily consumables across academic blocks, laboratories, hostels, dining messes, and administrative buildings. However, utility management on most campuses remains largely unmonitored or relies on static monthly billing cycles.

Critical operational inefficiencies—such as unoccupied lecture halls with active HVAC/lighting, unaddressed underground water pipeline leakages, unsegregated mess waste, and suboptimal solar grid utilization—go undetected in real time.

The objective of **PS-03 (Sustainable Campus & Facility Intelligence Dashboard)** is to design and develop an IoT-integrated, AI-driven energy and resource monitoring platform. The system must ingest live or simulated sensor telemetry across electrical grids, water reservoirs, and waste collection points to identify consumption anomalies, predict demand peaks, suggest automated conservation policies, and provide institutional leadership with transparent carbon-footprint accounting.

2. Core Personas & Access Roles

1. **Campus Estate & Facility Manager:** Monitors live telemetry across campus substations, main water pumps, solar inverter outputs, and waste collection checkpoints; manages maintenance repair tickets.
2. **Institutional Leadership / Management:** Reviews macro ESG (Environmental, Social, and Governance) compliance metrics, carbon-neutrality progress indicators, utility expenditure forecasts, and budget allocations.
3. **Department Heads / Hostel Wardens:** Accesses micro-level resource diagnostics for specific academic blocks or residential wings to enforce localized sustainability targets.
4. **Student & Campus Community (Public View):** Views interactive sustainability leaderboards, eco-bounty incentives for energy-saving, and live campus green-energy generation metrics to foster civic conservation habits.

3. Key Functional Modules & Technical Deliverables

Module 1: Real-Time Smart Energy Telemetry & Anomaly Engine

- **Multizone Power Monitoring:** Ingests real-time or simulated MQTT/HTTP sensor telemetry (voltage, current, power factor, kWh) segmented by building, laboratory, and residential floor.
- **Idle-Load & Phantom Drain Detection:** Algorithmic identification of active power draw in scheduled unoccupied zones (e.g., air conditioners or lab computers running post-curfew).
- **Renewable Energy Integration:** Live tracking of rooftop solar panel generation versus grid import, calculating self-sufficiency ratios.

Module 2: Water Network Monitoring & Leak Detection

- **Tank & Reservoir Level Telemetry:** Automated tracking of overhead and underground water storage with predictive pump automation triggers.
- **Pipeline Loss & Flow Anomaly Analytics:** Differential flow metering between supply pumps and destination manifolds to detect concealed pipe bursts and faucet leakages.
- **Rainwater Harvesting & Ground Recharge Metrics:** Inflow computation and recharge efficiency indicators during rainfall events.

Module 3: Food & Solid Waste Intelligence

- **Mess Waste Tracking:** Daily quantification of organic dining hall waste correlated against student attendance from mess logs.
- **Compost & Biogas Yield Predictor:** Mathematical modeling estimating bio-methane or organic fertilizer output potential from collected campus waste.
- **Smart Bin Fill-Level Optimization:** Route-efficiency alerts for campus sanitation workers based on ultrasonic bin fill sensors.

Module 4: Predictive Forecasting & Actionable Optimization AI

- **Demand Peak Forecasting:** Time-series ML model predicting electrical peak load requirements based on historical trends, weather feeds, and institutional academic calendars.
- **Prescriptive Efficiency Recommendations:** AI-generated automated operational interventions (e.g., *"Shift water pump cycle from 2:00 PM to 11:00 AM to consume surplus rooftop solar generation"*).
- **Audit-Ready Carbon Accounting:** Automated conversion of electrical and fossil usage into metric tons of \$CO_2\$ equivalent (\$MTCO_2e\$) aligned with national sustainability benchmarks.

4. Recommended Technology Stack

Layer, Suggested Technologies

Frontend Web, React.js (Vite) / Next.js / Tailwind CSS / Recharts / Chart.js / D3.js

Backend & Ingestion, Node.js (Express) or Python (FastAPI) with MQTT Broker (Mosquitto / HiveMQ)

Database & Telemetry Storage, InfluxDB / TimescaleDB (Time-series) + PostgreSQL / MongoDB / Firebase

Data Science & AI / ML, "Python (Pandas, Scikit-learn, Prophet / ARIMA, TensorFlow / PyTorch), Gemini API for prescriptive audit report summaries"

IoT / Hardware Simulation, "Wokwi / ESP32 Simulator, Node-RED, or Mock Data Generator scripts streaming realistic campus telemetry"

Authentication & RBAC, "JWT, OAuth2, Role-Based Access Control"

5. Evaluation & Judging Criteria (100 Points Total)

1. **Telemetry Architecture & Simulation Integrity (25 Points):** Robustness of the data ingestion pipeline handling time-series feeds for power, water, and waste metrics.
2. **Anomaly Detection & Predictive Utility (25 Points):** Efficacy in identifying real leaks, idle energy drains, and peak load forecasting rather than displaying static data graphs.
3. **Actionable Conservation & Sustainability Logic (20 Points):** Real utility of automated recommendations, solar load-shifting rules, and carbon-accounting formulas.
4. **UI/UX, Dashboards & Community Engagement (20 Points):** Clarity of administrative visualization, spatial heatmaps of campus blocks, and eco-leaderboard usability.
5. **Sprint Prototype Execution & Code Standards (10 Points):** Clean repository architecture, Dockerized or easily runnable local environment, and live deployment demonstration.

6. Submission Guidelines

- **Code Repository:** Public/Private GitHub repository with structured commits, .env.example, mock telemetry generator script, and clear setup documentation.
- **Live Deployment:** Working live web interface hosted on Vercel, Render, AWS, or Netlify.
- **Demo Video / Deck:** A concise 3-minute video highlighting live sensor data ingestion, anomaly triggering (e.g., simulated water leak or overnight power spike), and administrative carbon auditing.